The Case for Voluntary Disclosure: Strategic Feature Revelation in Regression

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Abstract

This paper addresses a fundamental design question: should a predictive model enforce mandatory feature disclosure or allow voluntary disclosure by participants? Since participants strategically reveal only those features that align with their private objectives, the model must account for opportunistic reporting. For the regression problem, we establish conditions under which the predictive model actually achieves better performance by permitting selective revelation.

1. INTRODUCTION

The performance of any predictive model depends on the set of features provided for prediction. In standard settings, all relevant features are assumed to be fully observed by the model. In this paper, we study a setting in which a participant can voluntarily choose which features to disclose. It is natural to assume that the participant will disclose only those features that are beneficial to their personal objective. The central question we address is: under what conditions does voluntary feature disclosure actually improve predictive performance?

For a predictive model to benefit from such disclosure, its inference procedure must account for the possibility of opportunistic reporting. This problem becomes intricate when participants do not share a uniform directional preference over outcomes, as these heterogeneous preferences make it difficult to exploit systematic patterns in feature withholding.

A concrete and illustrative example arises in financial markets when estimating a firm's valuation. The market may enforce a mandatory disclosure rule for financial information. If full disclosure is required, the resulting valuation achieves a baseline accuracy dependent on the complexity of the underlying relationship. Alternatively, the market might allow firms to voluntarily disclose financial details, in which case the valuation is formed based only on the revealed information.

Firms typically hold privately preferred valuations that differ from the market's assessment, and the preferred direction of this deviation is not uniform across firms. Some firms actively seek overvaluation because it raises stock prices and lowers the cost of equity financing. Others may prefer undervaluation, for example, when preparing for a stock buyback program or seeking to avoid triggering regulatory scrutiny.

If a firm's preferred valuation is closely aligned with the market's objective, selective feature disclosure may assist by intuitively filtering out noisy or less informative signals, thereby enhancing valuation accuracy relative to mandatory disclosure. In contrast, if the firm's preferences are misaligned with the market's objective, selective disclosure distorts the available information and introduces a systematic bias, thereby degrading the valuation accuracy.

In the speical case if firms shared the same directional preference—for instance, if every firm preferred overvaluation—the market could prevent exploitation simply by assigning a conservatively low default valuation in cases of partial disclosure. Such a rule would effectively discourage withholding, thereby replicating the outcome of a mandatory disclosure regime. In this work, however, we accommodate the more general and realistic setting in which participants hold heterogeneous preferences. We derive conditions to identify when the strategic disclosure of features is beneficial to the predictive model.

1.1. Main Contributions

In our work, we consider the predictive problem of regression, modeling the strategic interaction as a sender-receiver game.

Assuming the environment is fully known to both players, we establish sufficient conditions in the regression setting for the predictive model to benefit from voluntary disclosure. These conditions depend on the misalignment between the sender's preferred value and

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the ground truth value. We also provide necessary conditions that depend on the relationship between the disclosed features and the ground truth.

1.2. Related Works

Our work builds on the intersection of strategic machine learning, sender-receiver games, and voluntary disclosure in predictive modeling. Early studies in strategic machine learning, such as [1], formalized settings where agents manipulate features to influence outcomes, showing that robust models can mitigate gaming in linear settings. This was extended in [2] to dark (unknown) environments, where agents' responses are not fully observable, leading to equilibrium analyses under uncertainty.

Sender-receiver games provide a foundational framework for our analysis. [3] examined learning dynamics in these games using experimental data, highlighting how repeated interactions evolve into efficient signaling. Evolutionary perspectives, as in [4], explore multi-agent signaling, while [5] compared stimulus-response and belief-based learning frameworks.

Voluntary disclosure literature addresses selective revelation. [6] analyzed investor disclosures, predicting good-news bias, which parallels our heterogeneous preferences. Empirical studies, such as [7], used XGBoost and XAI for predicting disclosure patterns, emphasizing transparency in strategic contexts.

Stackelberg equilibria are central to our game-theoretic approach. [8] characterized equilibria in strategic regression, designing incentives to prevent cheating. [9] generalized PAC-learning for heterogeneous agents. In adversarial prediction, [10] framed problems as Stackelberg games, solving for optimal decision rules.

While these works advance strategic aspects in regression [11], they often assume known environments or linear models without noisy reporting channels. Our contributions provide formal sufficiency and necessity conditions under noisy channels and heterogeneous preferences, filling a gap in generalized predictive settings.

2. PROBLEM FORMULATION

Let $\mathscr{X} = \mathbb{R}^d$ be the feature space and $\mathscr{Y} = \mathbb{R}$ be the outcome space. We consider a data-generating process where Nature selects $(u,x,y) \sim (U,X,Y)$ over $\mathscr{Y} \times \mathscr{X} \times \mathscr{Y}$. Here, y is the ground truth value, x represents the available features, and u is the preferred value.

The problem is modeled as a game between two players: a **Sender** and a **Receiver**. The Sender observes (u,x) and chooses an action A from the strategy space:

$$\mathscr{S} = \{A : \mathscr{Y} \times \mathscr{X} \mapsto \{\emptyset, x\}\}.$$

This action represents the Sender's decision to either disclose the full feature vector x or withhold it (denoting the latter by $\emptyset$). The chosen action $A(u,x)$ is transmitted through a noisy channel, resulting in an observed message M. The channel noise is characterized by a parameter $q \in [0,1]$ such that:

$$M = \begin{cases} \emptyset & \text{if } A(u,x) = \emptyset, \\ \emptyset & \text{if } A(u,x) = x \text{ with probability } q, \\ x & \text{if } A(u,x) = x \text{ with probability } 1-q. \end{cases}$$

Upon observing $M \in \{\emptyset, x\}$, the Receiver applies a strategy $\hat{y} \in \mathscr{R}$:

$$\mathscr{R} = \{\hat{y} : \{\emptyset, \mathscr{X}\} \mapsto \mathscr{Y}\}.$$

The loss of each player is defined as follows: the Sender incurs a loss $\ell(u, \hat{y}(M))$ based on their preferred value u, while the Receiver incurs a loss $\ell(y, \hat{y}(M))$ based on the ground truth value y.

We analyze this interaction as a **screening game**. In this game, the Receiver first commits to a strategy $\hat{y} \in \mathscr{R}$. Then, the Sender observes this commitment and takes an optimal action $A \in \mathscr{S}$ based on $\hat{y}$. Finally, the Receiver executes their committed strategy and the outcome is realized.

For any commitment $\hat{y}$ by the Receiver, the Sender's **best response** is to choose an action $A(u,x)$ that minimizes their own expected loss. This reduces to a pointwise minimization for each observed state (u,x):

$$A(u,x) = \arg\min_{a \in \{\emptyset, x\}} \mathbb{E}\left[\ell(u, \hat{y}(M)) \mid A(u,x) = a\right]. \quad (1)$$

Anticipating the Sender's best response, the Receiver must now optimize the choice of $\hat{y}$ to minimize their own total expected loss, achieving the strategic risk R_S:

$$R_S := \min_{\hat{y} \in \mathscr{R}} \mathbb{E}[\ell(Y, \hat{y}(M))]. \quad (2)$$

Such a pair of strategies $(A, \hat{y})$ constitutes a **Stackelberg equilibrium**.

To evaluate the strategic gain, we compare this equilibrium to a mandatory disclosure regime (we also refer to this as the **vanilla** setting). Under mandatory disclosure, the Sender is forced to reveal features ($A \equiv x$), which leads to the Receiver's optimal strategy $\hat{y}_V$. The corresponding vanilla Risk is:

$$R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))]. \quad (3)$$

Definition 1 (Strategic Advantage)**.** *A **strategic advantage** exists if permitting voluntary disclosure yields a strictly lower expected loss for the Receiver than the mandatory regime, i.e., $R_S < R_V$.*

3. MAIN RESULTS

In this section, we characterize the conditions under which a strategic advantage arises. Throughout our analysis, we adopt the shorthand notation $\hat{y}(\emptyset)$ to denote the Receiver's prediction when features are withheld (also called default prediction), $\hat{y}(x)$ when features are revealed, $\hat{y}(M)$ when evaluating on the observed message M, and $\hat{y}_V(M)$ for the vanilla predictor's output on M.

We analyze the regression problem under the assumption of a known environment, deriving distinct necessary and sufficient conditions for the emergence of a strategic advantage. Detailed proofs for all results in this section are deferred to the Postponed Proofs section 5.

3.1. Strategic Advantage Condition in Regression

The outcome space $\mathscr{Y} = \mathbb{R}$ is the space of real numbers and the loss function $\ell(y, \hat{y}(M)) = (y - \hat{y}(M))^2$ is mean squared error (MSE). For this regression formulation, we explicitly assume that all random variables (U, X, Y) have zero mean everywhere. In the mandatory disclosure setup, the Sender always chooses $A = x$. For a linear Receiver strategy $\hat{y}_V(\emptyset) = 0$ and $\hat{y}_V(x) = b^\top x$, we characterize the resulting risk in the following lemma.

Lemma 1. *In the mandatory disclosure setup, for a Receiver strategy parameterized by $b \in \mathbb{R}^d$, the expected risk is*

$$R_V(b,q) = \mathrm{Var}(Y) - (1-q)\,\mathrm{Var}(b_Y^\top X) + (1-q)\,\mathrm{Var}((b - b_Y)^\top X), \quad (4)$$

where $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X,Y)$. The optimal vanilla risk is achieved at $b = b_Y$, yielding

$$R_V^\star(q) = \mathrm{Var}(Y) - (1-q)\,\mathrm{Var}(b_Y^\top X). \quad (5)$$

Remark 1. *Note that $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X,Y)$ is the optimal weight vector for ordinary least squares (OLS) regression of Y on X.*

In the voluntary disclosure setup, for a given Receiver strategy $\hat{y}(M)$, we consider the linear form $\hat{y}(\emptyset) = 0$ and $\hat{y}(x) = b^\top X$, where $b \in \mathbb{R}^d$. For a fixed b, we define the following scalar fields:

$$F(x,u) := (2u - b^\top x)b^\top x, \quad (6)$$

$$G(x,y) := (2y - b^\top x)b^\top x. \quad (7)$$

The Sender's best response action is then characterized by the following lemma.

Lemma 2. *In the voluntary disclosure setup, for a given receiver strategy $\hat{y}(M)$, the Sender's best response action is given by*

$$A(x,u) = x \iff F(x,u) > 0. \quad (8)$$

Based on this best response, we derive the Receiver's strategic risk R_S in the next lemma.

Lemma 3. *Given a Receiver strategy $\hat{y}(M)$ parameterized by b, the resulting strategic risk is*

$$R_S(b,q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}[G(X,Y)\mathbb{I}\{F(X,U) > 0\}], \quad (9)$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function.

Remark 2. *The Stackelberg equilibrium risk is defined as the minimum strategic risk over the parameter space:*

$$R_S^\star(q) := \min_b R_S(b,q).$$

Remark 3. *The function $F(x,u)$ serves as the decision rule for the Sender. The function $G(x,y)$ represents the Receiver's gain when the feature vector x is successfully received through the noisy channel (i.e., the signal is not lost to noise) relative to when the feature vector is lost.*

The strategic advantage condition is $R_V^\star(q) - R_S^\star(q) > 0$. We provide a sufficiency condition for this inequality in the following theorem.

Theorem 1 (Sufficiency Condition)**.** *A strategic advantage exists if there exists $b \in \mathbb{R}^d$ such that:*

$$\|U - Y\|_2 < \frac{\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}} - \frac{\mathrm{Var}((b - b_Y)^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}}, \quad (10)$$

where $\|U - Y\|_2 = \sqrt{\mathrm{Var}(U) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(U,Y)}$, $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X,Y)$, and $G(X,Y) = (2Y - b^\top X)b^\top X$.

Beyond sufficiency, we can also identify a necessary condition for a strategic advantage to be possible. We define a strategic disadvantage when $R_S^\star(q) = R_V^\star(q)$; therefore, a necessary condition for an advantage must be the complement of this state.

Theorem 2 (Necessary Condition)**.** *A necessary condition for a strategic advantage to exist is:*

$$\max_b \Big(\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X)\Big) > 0. \quad (11)$$

Remark 4. *The necessary condition in* (11) *is equivalent to the requirement that the sufficiency condition in* (10) *be non-vacuous. Since the left-hand side of* (10) *is non-negative, the condition in* (11) *is equivalent to the numerator on the right-hand side being positive (as the denominator is always positive), thereby allowing for the existence of a non-empty range for* $\|U - Y\|_2$.

Remark 5. *Both the necessary and sufficient conditions are independent of the channel noise parameter q. This highlights that channel noise only controls the magnitude of the strategic advantage, not whether it can be attained. Additionally, both theorems require the bounds to hold for the best possible parameter b. Consequently, if the conditions are satisfied evaluated simply at the vanilla optimal weight* b_Y*, they necessarily hold under optimal b.*

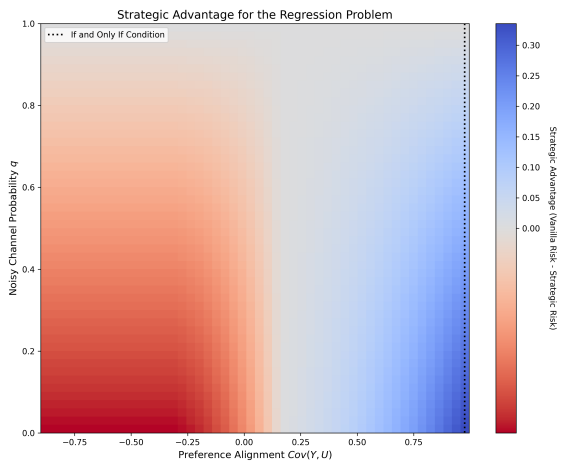

Figure 1. Heatmap of the strategic advantage ($R_V^\star(q) - R_S^\star(q) > 0$) for the general regression setup. The parameter space sweeps the noisy channel probability q along the y-axis and the preference alignment $\mathrm{Cov}(Y,U)$ along the x-axis. The dotted black contour delineates the region where the sufficiency condition is met.

To numerically verify these theoretical results, we simulate a regression environment using a family of zero-mean Gaussian distributions for (X,Y,U) with unit marginal variances. We fix $\mathrm{Cov}(X,Y) = \mathrm{Cov}(X,U) = 0.2$ across this family, while $\mathrm{Cov}(Y,U)$ serves as the varying parameter. To explore the parameter space, we sweep the channel noise probability $q \in [0,1]$ and the alignment $\mathrm{Cov}(Y,U) \in [-1,1]$. At each point on the grid, the Receiver's risk is minimized over a range of weights b to calculate the strategic risk R_S. As demonstrated in Fig. 1, the analytical boundary reliably delineates the regions of strategic advantage.

4. CONCLUSION

In this paper, we characterized the conditions under which a predictive model benefits from permitting voluntary feature disclosure rather than enforcing mandatory disclosure. Focusing on the regression setting, we framed the strategic interaction as a sender-receiver game and established formal sufficiency and necessity conditions for the existence of a strategic advantage. These conditions highlight the critical role of preference misalignment and the impact of the noisy transmission channel parameter q on the receiver's optimal disclosure regime design.

4.1. Limitations

Our current analysis operates under certain assumptions that limit its generality. First, the derived sufficiency bounds, while theoretically sound, can be conservative and may not tightly capture the exact transition point where voluntary disclosure becomes beneficial in all distribution families. Second, the theoretical framework currently restricts the receiver to linear strategies. While this choice provides tractability and clean, interpretable algebraic conditions, it restricts the receiver from deploying highly expressive, non-linear predictors that could potentially extract more nuanced information from opportunistic reporting.

4.2. Future Work

There are several promising directions for extending this research. A natural next step is to adapt these bounds beyond continuous regression to handle discrete problem formulations, such as binary classification, where the interplay of conditional distributions may lead to different threshold behaviors. Furthermore, our analysis assumes the environment and distributions are perfectly known to the receiver. Extending our framework to an online learning or dynamic setting—where the receiver must interact repeatedly with senders to learn the environment parameters while minimizing cumulative regret—remains a compelling direction. Finally, extending the game-theoretic formulation to accommodate multi-sender environments, where strategic feature disclosure from one entity influences the competitive dynamics of others, represents a challenging but highly applicable frontier.

5. POSTPONED PROOFS

Here we present the proof of all results we skipped in the Section 3.

5.1. Proofs for Strategic Advantage Condition for Regression

This section provides the complete derivations for the general regression risk formulations and the subsequent sufficiency and necessity conditions.

Proof of Lemma 1. We compute the risk in the mandatory disclosure setting for a linear predictor $b \in \mathbb{R}^d$ with $\hat{y}_V(\emptyset) = 0$ and $\hat{y}_V(x) = b^\top x$. The risk is given by:

$$R_V(b,q) = q\mathbb{E}[Y^2] + (1-q)\mathbb{E}[(Y - b^\top X)^2].$$

Since all variables have mean 0, $\mathrm{Var}(Y) = \mathbb{E}[Y^2]$. We now examine the term $\mathbb{E}[(Y - b^\top X)^2]$. We add and subtract $b_Y^\top X$ inside the expectation, where $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X,Y)$:

$$\begin{aligned}\mathbb{E}[(Y - b^\top X)^2] &= \mathbb{E}[(Y - b_Y^\top X)^2] \\ &\quad + \mathbb{E}[((b_Y - b)^\top X)^2] \\ &\quad + 2\mathbb{E}[(Y - b_Y^\top X)(b_Y - b)^\top X].\end{aligned}$$

By the orthogonality principle of OLS, $\mathbb{E}[(Y - b_Y^\top X)X] = 0$, so the cross-term vanishes. Thus,

$$\mathbb{E}[(Y - b^\top X)^2] = \mathrm{Var}(Y) - \mathrm{Var}(b_Y^\top X) + \mathrm{Var}((b - b_Y)^\top X).$$

Substituting back into the risk yields (4). The optimal strategy minimizes $\mathrm{Var}((b - b_Y)^\top X)$ by choosing $b = b_Y$. □

Proof of Lemma 2. Given a realization (x,u), the Sender selects the action $A \in \{\emptyset, x\}$ minimizing $(u - \hat{y}(M))^2$. The expected loss from revealing is $qu^2 + (1-q)(u - b^\top x)^2$, while from withholding it is u^2. The Sender prefers revealing if:

$$u^2 > qu^2 + (1-q)(u - b^\top x)^2.$$

Simplifying yields $(b^\top x)(2u - b^\top x) > 0$, which is $F(x,u) > 0$. □

Proof of Lemma 3. For a fixed parameter b, the strategy of the Receiver is $\hat{y}_V(\emptyset) = 0$ for an empty message $M = \emptyset$ and $\hat{y}_V(x) = b^\top x$ for a successfully received feature vector $M = X$.

According to Lemma 2, the Sender reveals the feature vector x if and only if $F(X,U) > 0$. Using the law of total expectation, we decompose the strategic risk $R_S(b,q) = \mathbb{E}[\ell(Y, \hat{y}(M))]$ into the two cases of F:

$$\begin{aligned}R_S(b,q) &= \mathbb{E}\big[\mathbb{I}\{F \le 0\}\ell(Y, \hat{y}(\emptyset)) \\ &\quad + \mathbb{I}\{F > 0\}(q\ell(Y, \hat{y}(\emptyset)) + (1-q)\ell(Y, \hat{y}(X)))\big] \\ &= \mathbb{E}\big[\mathbb{I}\{F \le 0\}Y^2 \\ &\quad + \mathbb{I}\{F > 0\}(qY^2 + (1-q)(Y - b^\top X)^2)\big].\end{aligned}$$

Applying the identity $\mathbb{I}\{F \le 0\} = 1 - \mathbb{I}\{F > 0\}$:

$$\begin{aligned}R_S(b,q) &= \mathbb{E}[Y^2] - (1-q)\mathbb{E}\big[\mathbb{I}\{F > 0\}(Y^2 - (Y - b^\top X)^2)\big] \\ &= \mathrm{Var}(Y) - (1-q)\mathbb{E}\big[\mathbb{I}\{F > 0\}(2Y(b^\top X) \\ &\quad - (b^\top X)^2)\big],\end{aligned}$$

where $\mathrm{Var}(Y) = \mathbb{E}[Y^2]$ follows from the zero-mean assumption. Substituting $G(X,Y) = (2Y - b^\top X)b^\top X$ yields:

$$R_S(b,q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}[\mathbb{I}\{F > 0\}G(X,Y)],$$

which concludes the proof. □

Proof of Theorem 1 (Sufficiency Condition). We seek condition such that $R_S^\star(q) - R_V^\star(q) < 0$. From Lemmas 1 and 3:

$$\begin{aligned}R_S^\star(q) - R_V^\star(q) &= \mathrm{Var}(Y) - (1-q)\mathbb{E}[G\mathbb{I}\{F > 0\}] \\ &\quad - \big(\mathrm{Var}(Y) - (1-q)\,\mathrm{Var}(b_Y^\top X)\big).\end{aligned}$$

Using $\mathbb{I}\{F > 0\} = 1 - \mathbb{I}\{F \le 0\}$, we have $\mathbb{E}[G\mathbb{I}\{F > 0\}] = \mathbb{E}[G] - \mathbb{E}[G\mathbb{I}\{F \le 0\}]$.

To compute $\mathbb{E}[G]$, recall $G(X,Y) = (2Y - b^\top X)b^\top X$. Since $\mathbb{E}[X] = 0$, $\mathrm{Var}(b^\top X) = \mathbb{E}[(b^\top X)^2]$. Furthermore, the optimal OLS weights $b_Y = \mathbb{E}[XX^\top]^{-1}\mathbb{E}[XY]$ satisfy $\mathbb{E}[Y(b^\top X)] = b^\top \mathbb{E}[XY] = b^\top \mathrm{Var}(X)b_Y = \mathrm{Cov}(b^\top X, b_Y^\top X)$. Thus,

$$\begin{aligned}\mathbb{E}[G] &= 2\mathbb{E}[Y(b^\top X)] - \mathbb{E}[(b^\top X)^2] \\ &= 2b^\top \mathrm{Var}(X)b_Y - \mathrm{Var}(b^\top X).\end{aligned}$$

From the identity $\mathrm{Var}((b - b_Y)^\top X) = \mathrm{Var}(b^\top X) + \mathrm{Var}(b_Y^\top X) - 2b^\top \mathrm{Var}(X)b_Y$, we obtain:

$$\mathbb{E}[G] = \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X).$$

Substituting this into the risk difference we get:

$$R_S^\star(q) - R_V^\star(q) = (1-q)\big(\mathrm{Var}((b - b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{F \le 0\}]\big).$$

We upper bound $\mathbb{E}[G\mathbb{I}\{F \le 0\}]$ by decomposing:

$$\begin{aligned}\mathbb{E}[G\mathbb{I}\{F \le 0\}] &= \mathbb{E}[G\mathbb{I}\{G \le 0\}] \\ &\quad + \mathbb{E}[G(\mathbb{I}\{F \le 0\} - \mathbb{I}\{G \le 0\})] \\ &= -\tfrac{1}{2}\big(\mathbb{E}[|G|] - \mathbb{E}[G]\big) \\ &\quad + \mathbb{E}[G\mathbb{I}\{\mathrm{sign}(F) \ne \mathrm{sign}(G)\}].\end{aligned}$$

When $\text{sign}(F) \neq \text{sign}(G)$, we have $|G| \leq |G-F| = 2|Y-U||b^\top X|$. By Cauchy-Schwarz:

$$\mathbb{E}[G\mathbb{I}\{\text{sign}(F) \neq \text{sign}(G)\}] \leq \mathbb{E}[2|Y-U|\,|b^\top X|] \leq 2\|U-Y\|_2\sqrt{\text{Var}(b_Y^\top X)}.$$

Setting upper bound to be negative, (which would imply $R_S^\star(q) - R_V^\star(q) < 0$), we get:

$$\begin{aligned}&\text{Var}((b-b_Y)^\top X)\\ &-\tfrac{1}{2}\big(\mathbb{E}[|G|] - \text{Var}(b_Y^\top X)\\ &\quad+\text{Var}((b-b_Y)^\top X)\big)\\ &+2\|U-Y\|_2\sqrt{\text{Var}(b_Y^\top X)} < 0.\end{aligned}$$

Rearranging isolates $\|U-Y\|_2$, yielding (10). □

Proof of Theorem 2 (Necessary Condition). We prove the contrapositive: if (11) fails, then no strategic advantage can exist, i.e., $R_S(b,q) \geq R_V^\star(q)$ for all b. From Lemma 1 and Lemma 3, we have:

$$R_S(b,q) - R_V^\star(q) = (1-q)\big(\text{Var}((b-b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{F \leq 0\}]\big).$$

A strategic advantage requires $R_S(b,q) - R_V^\star(q) < 0$ for some b. We can lower bound the term $\mathbb{E}[G\mathbb{I}\{F \leq 0\}]$ purely algebraically.

If $G \leq 0$, we have $G\mathbb{I}\{F \leq 0\} \geq G = G\mathbb{I}\{G \leq 0\}$. If $G > 0$, we have $G\mathbb{I}\{F \leq 0\} \geq 0 = G\mathbb{I}\{G \leq 0\}$. Therefore, pointwise $G\mathbb{I}\{F \leq 0\} \geq G\mathbb{I}\{G \leq 0\}$, which implies:

$$\mathbb{E}[G\mathbb{I}\{F \leq 0\}] \geq \mathbb{E}[G\mathbb{I}\{G \leq 0\}].$$

Substituting this inequality yields:

$$R_S(b,q) - R_V^\star(q) \geq (1-q)\big(\text{Var}((b-b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{G \leq 0\}]\big).$$

Now $\mathbb{E}[G\mathbb{I}\{G \leq 0\}]$ depends only on (X,Y) and b, with U entirely eliminated. Using the identity $\mathbb{E}[G\mathbb{I}\{G \leq 0\}] = -\frac{1}{2}(\mathbb{E}[|G|] - \mathbb{E}[G])$ and $\mathbb{E}[G] = \text{Var}(b_Y^\top X) - \text{Var}((b-b_Y)^\top X)$:

$$\begin{aligned}R_S(b,q) - R_V^\star(q) \geq (1-q)\Big(&\text{Var}((b-b_Y)^\top X)\\ &-\tfrac{1}{2}\big(\mathbb{E}[|G|] - \text{Var}(b_Y^\top X)\\ &+\text{Var}((b-b_Y)^\top X)\big)\Big).\end{aligned}$$

Multiplying by 2 and rearranging, $R_S(b,q) - R_V^\star(q) < 0$ requires:

$$\mathbb{E}[|G|] - \text{Var}(b_Y^\top X) - \text{Var}((b-b_Y)^\top X) > 0.$$

If this fails for all b, then $R_S(b,q) - R_V^\star(q) \geq 0$ for all b, meaning no strategic advantage is possible. Hence the necessary condition is that there exists some b satisfying (11) and thus

$$\max_b \mathbb{E}[|G|] - \text{Var}(b_Y^\top X) - \text{Var}((b-b_Y)^\top X) > 0.$$

□

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